

$$3) \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = x^2 - 3$$

$$r^2 - 3r + 2 = 0 \quad r_1 = 2 \quad r_2 = 1$$

$$(D^2 - 3D + 2)y = x^2 - 3$$

$$y = \frac{1}{D^2 - 3D + 2} (x^2 - 3)$$

$$y = \left(\frac{1}{2} + \frac{3}{4}D + \frac{7}{8}D^2 \right) (x^2 - 3)$$

$$y = \frac{x^2}{2} - \frac{3}{2} + \frac{3}{4}(2x) + \frac{7}{8} \cdot 2$$

$$y = \frac{x^2}{2} - \frac{3}{2} + \frac{6x}{4} + \frac{14}{8}$$

$$y = \frac{x^2}{2} + \frac{3x}{2} + \frac{1}{4}$$

$$\begin{array}{r} 1 \\ - 1 + \frac{3}{2}D - \frac{D^2}{2} \\ \hline \frac{1}{2} + \frac{3}{4}D + \frac{7}{8}D^2 \\ \hline - \frac{3}{2}D + \frac{9}{4}D^2 + \frac{3}{4}D^3 \\ \hline \frac{7}{4}D^2 + \frac{3}{4}D^3 \\ \hline - \frac{7}{4}D^2 + \frac{21}{8}D^3 - \frac{7}{8}D^4 \\ \hline \end{array}$$

Cuarto Caso: $B(x)$ es de la forma $e^{ax} \cdot F(x)$

$$P(D)y = e^{ax}F(x) \rightarrow y_p = e^{ax} \cdot G(x)$$

$$y_p = e^{ax} \left[\frac{1}{P(D+a)} F(x) \right] \text{ Toda vez que } P(D+a) \neq 0$$

$$y_p = e^{ax} \left[\frac{1}{P(D+a)} F(x) \right]$$

$$4) \frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = e^x \cdot \sin 2x$$

$$(D^2 - 3D + 2)y = e^x \cdot \sin 2x$$

$$y = \frac{1}{D^2 - 3D + 2} (e^x \cdot \sin 2x) \quad a = 1$$

$$y = e^x \frac{1}{D^2 - 3(D+1) + (D+1)^2} \sin 2x$$

$$y = e^x \left[\frac{1}{\cancel{D^2 - 3D - 3} + D^2 + 2D + 1} \right] \sin 2x$$

$$y = e^x \left[\frac{1}{D^2 - D} \right] \sin 2x$$

